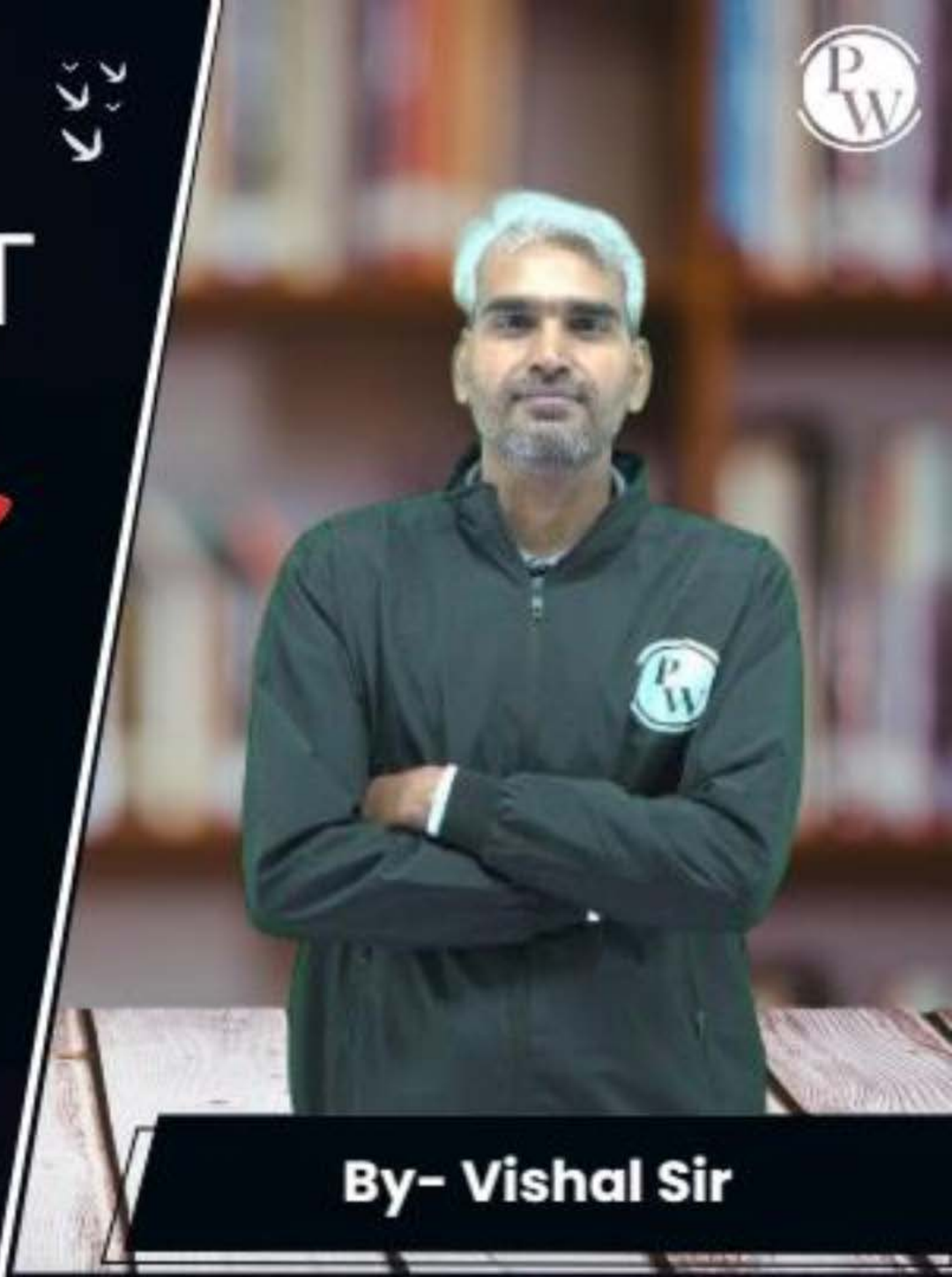
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



- Topic: Representation of graph
- Topic: Simple graph, multi graph and pseudo graph
- Topic: Maximum number of edges in a simple graph
- Topic: Maximum number of simple graph with n vertices
- Topic: Terminologies related to graphs

Slide

Topics to be Covered



- Topic: Degree of a vertex
- Topic: Null graph and Empty graph
- Topic: Different types of graphs

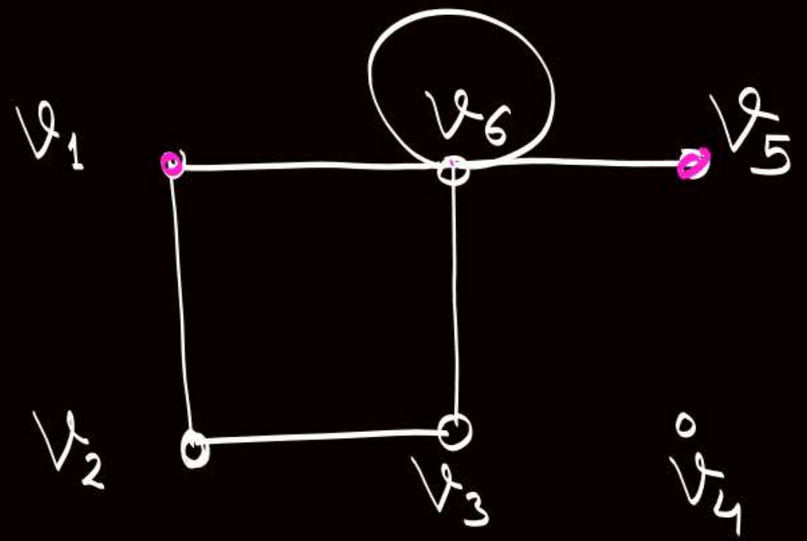
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Topic : Degree of a vertex in an Undirected Graph

- In an undirected graph degree of a vertex 'v' can be denoted by deg(v) and it is defined as no. of edges incident with vertex 'v'

> deg(v) = No. of edges incident with vertex 'v'



In a non-directed graph a loop at a vertex will be counted as two edges in order to obtain the degree of that vertex

$\deg(v_1) = 2$

$\deg(v_2) = 2$

$\deg(v_3) = 2$

$\deg(v_4) = 0$

$\deg(v_5) = 1$

$\deg(v_6) = 3 + 2 = 5$



Topic : Degree of a vertex in a directed Graph

- In a directed graph we define two types of degrees for a vertex

  1. In-degree
  2. Out degree



Topic : Degree of a vertex in a directed Graph

In-degree :- In-degree of a vertex 'v' is denoted by $deg^+(v)$ and it is defined as

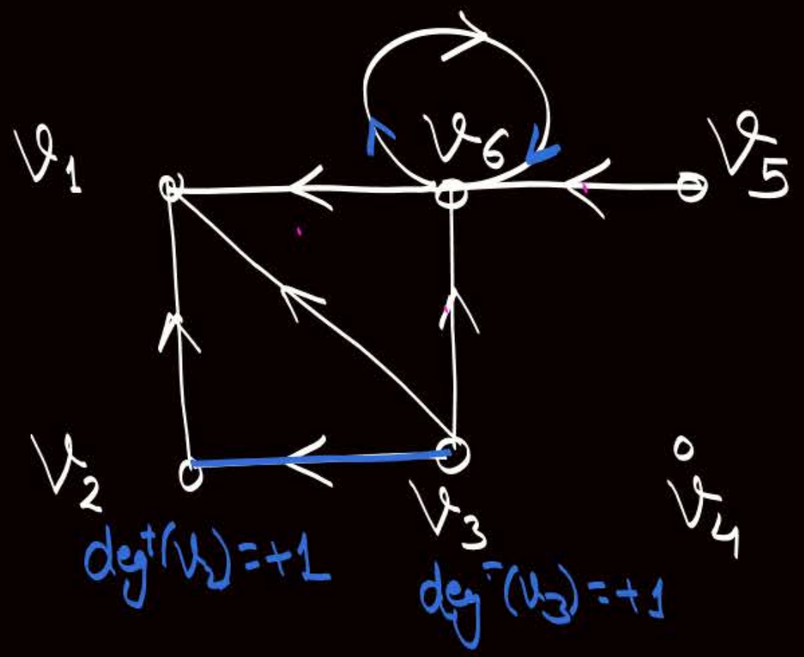
$deg^+(v)$ = No. of edges incoming to vertex v

Out-degree :- Out-degree of a vertex 'v' is denoted by $deg^-(v)$ and it is defined as

$deg^-(v)$ = No. of edges out going from vertex v

Summation of in-degree of vertex 'v' and out-degree of vertex 'v' will define the overall degree of vertex 'v'.

$$\text{i.e, } \deg(v) = \deg^{+}(v) + \deg^{-}(v)$$



In a directed graph a loop at a vertex will be counted as '1' edge to obtain the in-degree of that vertex and it will be counted as '1' edge to obtain the out-degree of that vertex

$\deg^+(v_1) = 3$ $\deg^+(v_2) = 1$ $\deg^+(v_3) = 0$ $\deg^+(v_4) = 0$ $\deg^+(v_5) = 0$ $\deg^+(v_6) = 2+1 = 3$

$\deg^-(v_1) = 0$ $\deg^-(v_2) = 1$ $\deg^-(v_3) = 3$ $\deg^-(v_4) = 0$ $\deg^-(v_5) = 1$ $\deg^-(v_6) = 1+1 = 2$

$\Sigma = 3$ $\Sigma = 2$ $\Sigma = 3$ $\Sigma = 0$ $\Sigma = 1$ $\Sigma = 5$

$\deg(v_1)$ $\deg(v_2)$ $\deg(v_3)$ $\deg(v_4)$ $\deg(v_5)$ $\deg(v_6)$

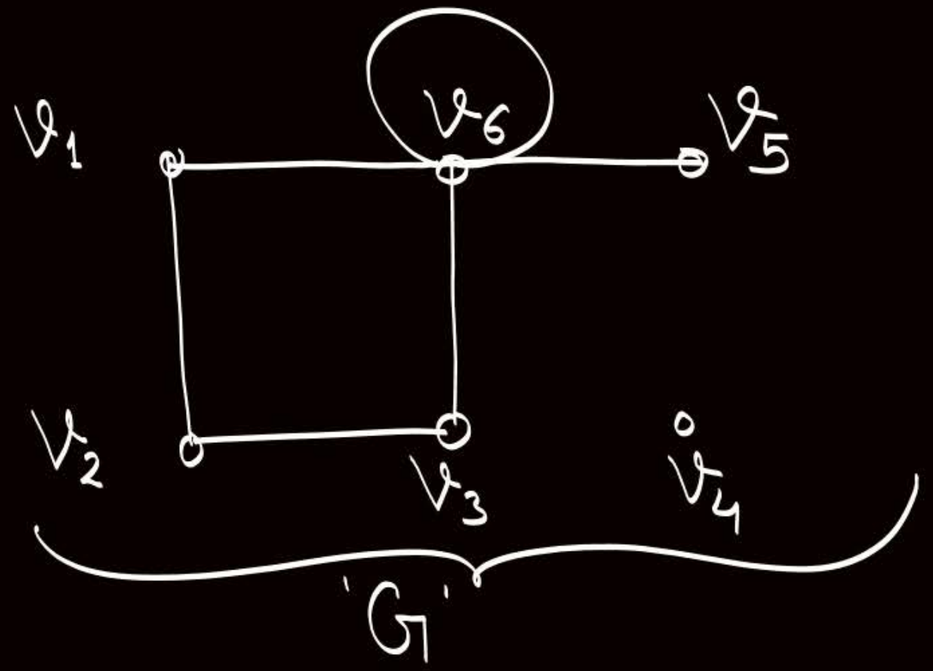


Topic : $\delta(G)$ and $\Delta(G)$

- In a graph G

$\delta(G)$: Minimum of the degrees of all the vertices of graph G is denoted by $\delta(G)$

$\Delta(G)$ = Maximum of the degrees of all the vertices of graph G is denoted by $\Delta(G)$



w.r.t. given given graph 'G'.

$\delta(G) = 0$

$\Delta(G) = 5$

$\deg(v_1) = 2$

$\deg(v_2) = 2$

$\deg(v_3) = 2$

$\deg(v_4) = 0$

$\deg(v_5) = 1$

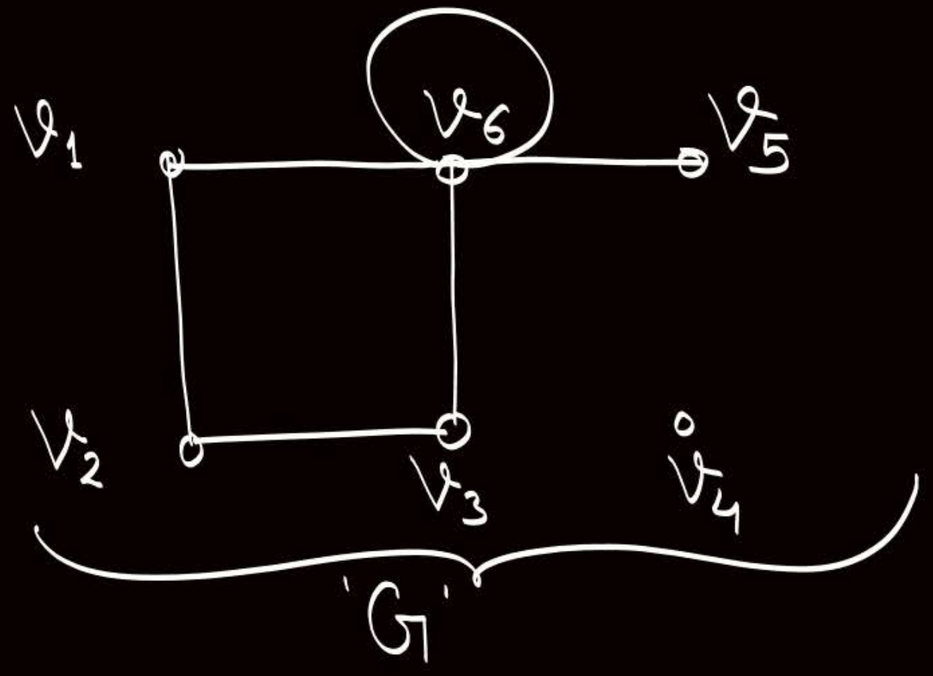
$\deg(v_6) = 3 + 2 = 5$



Topic : Isolated Vertex

In a graph G a vertex of degree = 0 is called an isolated vertex

i.e. if $\deg(v) = 0$, then 'v' is an isolated vertex



'v_4' is an isolated vertex

$\deg(v_1) = 2$

$\deg(v_2) = 2$

$\deg(v_3) = 2$

$\deg(v_4) = 0$

$\deg(v_5) = 1$

$\deg(v_6) = 3 + 2 = 5$



Topic : Null Graph

- A graph with no edges in it is called a NULL graph

{ vertices may be present in a NULL graph }

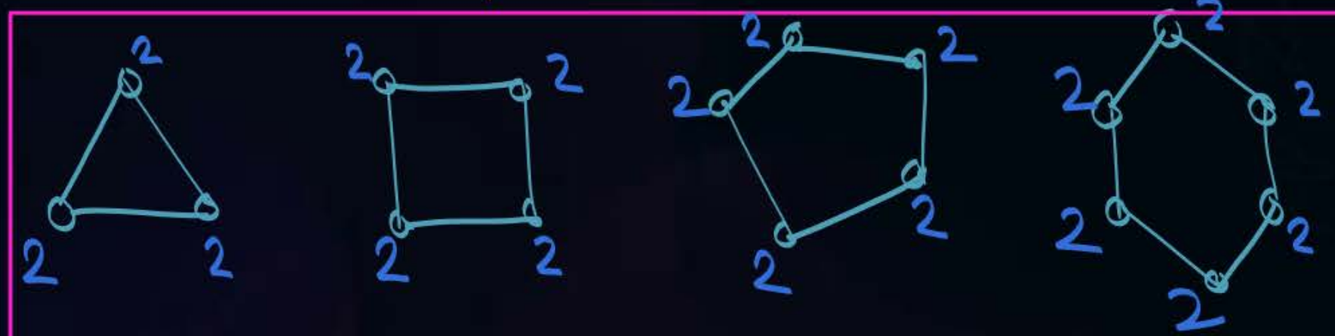
- **Empty Graph:-** A graph with no vertex and no edges is called an Empty graph



Topic : Regular Graphs

- In a graph G, if degree of each vertex is exactly same, then graph G is called a regular graph

- In a graph G, if degree of each vertex is exactly 'K' then graph G is called a 'K-regular' graph



Every Polygon is a '2-regular' graph



Topic : Types of graphs

1. Complete Graph (K_n)
2. Cycle Graph (C_n)
3. Wheel Graph (W_n)
4. Connected Graph
5. Cyclic Graph
6. Acyclic Graph
7. Tree
8. Bipartite Graph
9. Complete Bipartite Graph

Next Class



Topic : Complete Graph / K_n

- A Complete graph with n-vertices is denoted by 'K_n'
- K_n is a simple graph with n-vertices and all ${}^nC_2 = \frac{n(n-1)}{2}$ edges



* In a Complete graph K_n all n vertices are mutually adjacent

* In a Complete graph K_n degree of each vertex is exactly '$n-1$' $\therefore$ K_n is a '$n-1$' regular graph

* In a Complete graph K_n

$$|E(K_n)| = {}^{n}C_2 = \frac{n(n-1)}{2}$$

$|E(K_n)|$ → No. of Edges in Complete graph K_n



Topic : Cycle Graph

"C_n"

* A simple graph with n-vertices and n-edges ($n \geq 3$) where all n edges form a cycle of length 'n' is called a Cycle graph
* A cycle graph with n-vertices is denoted by 'C_n'
* Length of a cycle = No. of edges forming that cycle



C_3



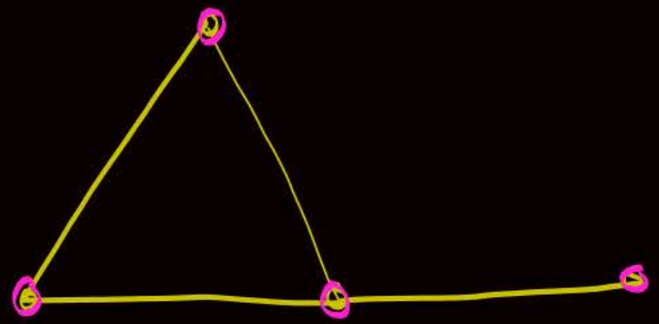
C_4



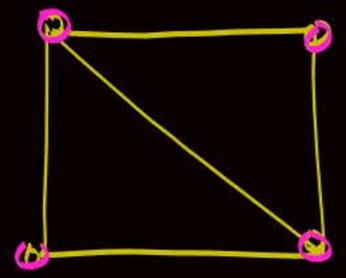
C_5

In a cycle graph C_n

$$|E(C_n)| = n$$



'4' Vertices
and '4' Edges
but it is not a
Cycle graph



No. of Vertices = 4
No. of Edges = 5
Not a Cycle graph



Topic : Wheel Graph

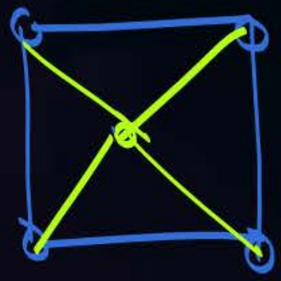
'W_n'

A wheel graph with n-vertices is denoted by W_n.

A wheel graph with n-vertices can be obtained by connecting a hub vertex with all the vertices of cycle graph C_{n-1} ($W_n = C_{n-1} + 1\text{-hub}$)



$W_4 = C_3 + 1\text{-hub}$



$W_5 = C_4 + 1\text{-hub}$



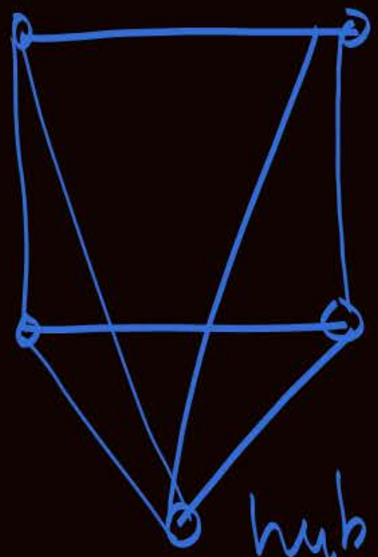
$W_6 = C_5 + 1\text{-hub}$

In a wheel graph W_n

$$|E(W_n)| = (n-1) + (n-1) = 2(n-1) = 2n-2$$

$(n-1)$: w.r.t. C_{n-1}

$(n-1)$: To connect hub vertex with $(n-1)$ vertices of C_{n-1}



W_5

In a wheel graph W_n

- degree of hub vertex = $(n-1)$
- degree of Every other vertex = 3 (Except hub vertex)

Every wheel graph W_n need not be a regular graph

- A wheel graph W_n is a regular graph if and only if $(n-1) = 3$

i.e. $\boxed{n = 4}$

The only wheel graph which is also a regular graph is W_4

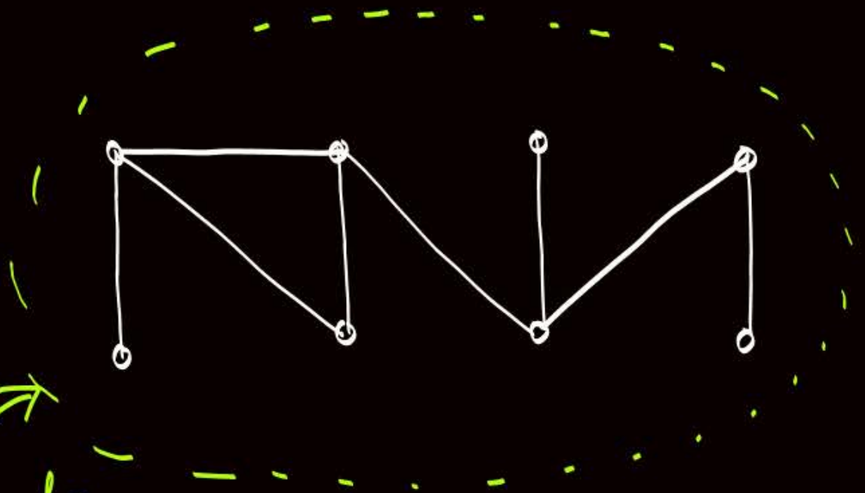


Topic : Connected Graph

- A graph with two or more Vertices is said to be connected if and only if there exist a path between Every pair of vertices

- A Single isolated Vertex may be treated differently in different Scenario.

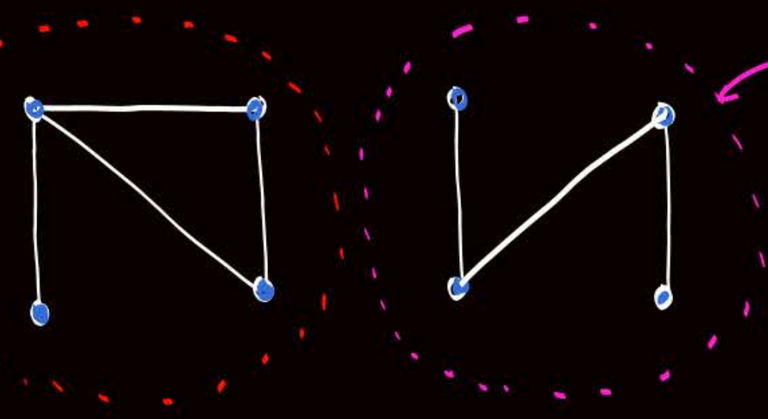
eg:



A single Connected Component.

graph 'G_1' is a connected graph + simple

It is a Connected Component



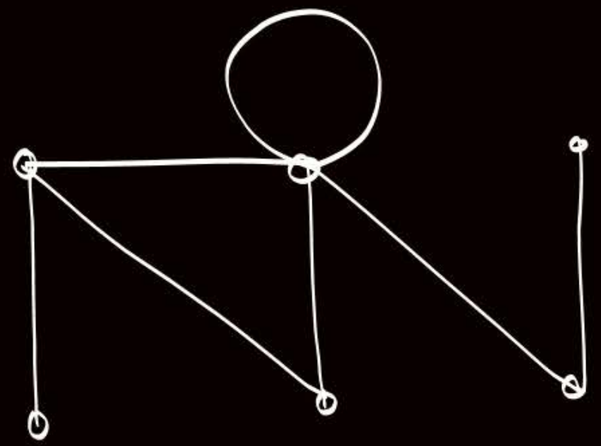
It is a Connected Component

graph 'G_2' is not a Connected graph + simple

In graph G_2 we have more than one Connected Components

- If graph is a Connected graph then there will be Exactly '1' Connected Component

- If graph is a disconnected graph then it will have two or more Connected Components.



It is a Connected graph
but it is not a simple graph

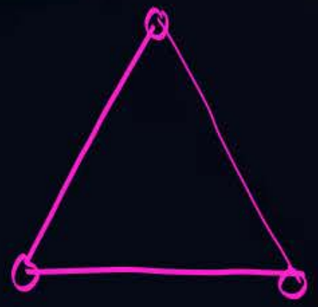
* A Connected graph may or may not be a simple graph

- K_n, C_n and W_n are always simple as well as Connected graph



Topic : Cyclic Graph

- A graph with at least one cycle in it is called a cyclic graph

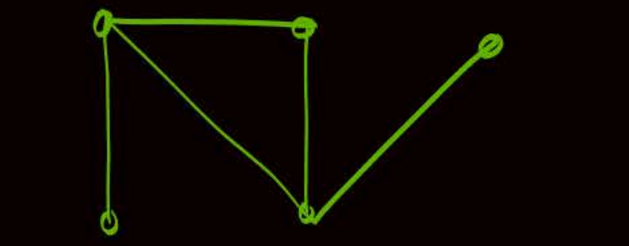


It is a cyclic graph as well as a cycle graph



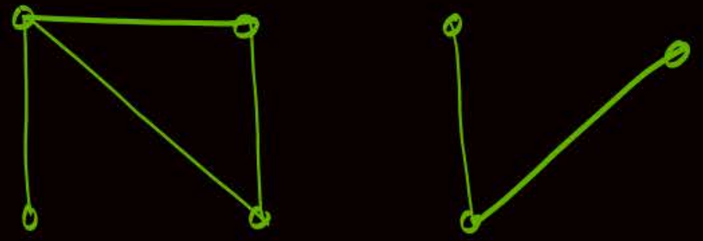
We have at least one cycle in it, ∴ it is a cyclic graph, but it is not a cycle graph

* Every cycle graph is also a cyclic graph,
but Every cyclic graph need not be a cycle graph



G_1

Cyclic graph
+
Connected graph



G_2

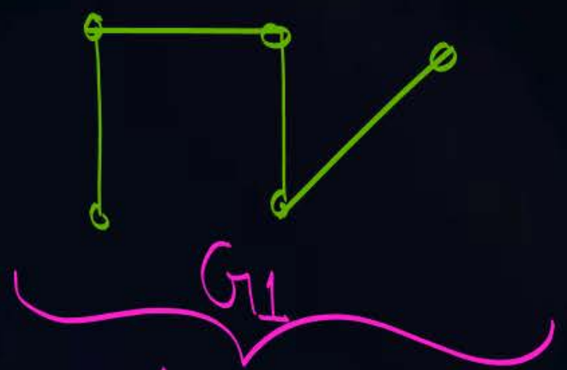
Cyclic graph
+
disconnected graph

* A cyclic graph may or may not be connected
* A cyclic graph may or may not be simple



Topic : Acyclic Graph

* A graph with no cycle in it is called an 'acyclic' graph.



G_1

Acyclic
+
Connected



G_2

Acyclic
+
disconnected

* Acyclic graph are always simple



Topic : Tree

- A tree is a Connected and acyclic graph

- A tree with n-vertices will have exactly 'n-1' edges

- A Collection of tree is Called a Forest
  - A Forest is Collection of 1 or more trees
  - A forest is a Collection of one or more acyclic Components

H.W.

Q: A Forest with N-vertices and 'K' Components will have how many edges ?

i.e 'K' trees in the Forest



2 mins Summary

- Topic: Degree of a vertex
- Topic: Null graph and Empty graph
- Topic: Different types of graphs

THANK - YOU

Slide